

1. Degree Distribution Under Configuration Model and Edge-Swapping

Dataset. Email-EU-core network (SNAP): a communication network from a European research institution loaded as an undirected simple graph. **986 nodes, 16,064 edges**, mean degree 32.6, max degree 345.

Configuration Model. Each node receives stubs equal to its degree; stubs are paired uniformly at random and self-loops / multi-edges are discarded. 100 instances were generated and degree distributions averaged.

Edge-Swapping. Starting from the original graph, two edges are selected at random and their endpoints swapped if the swap creates no self-loops or duplicate edges. $Q = 5$ accepted swaps per edge per instance (100 instances). The degree sequence is preserved exactly in every instance.

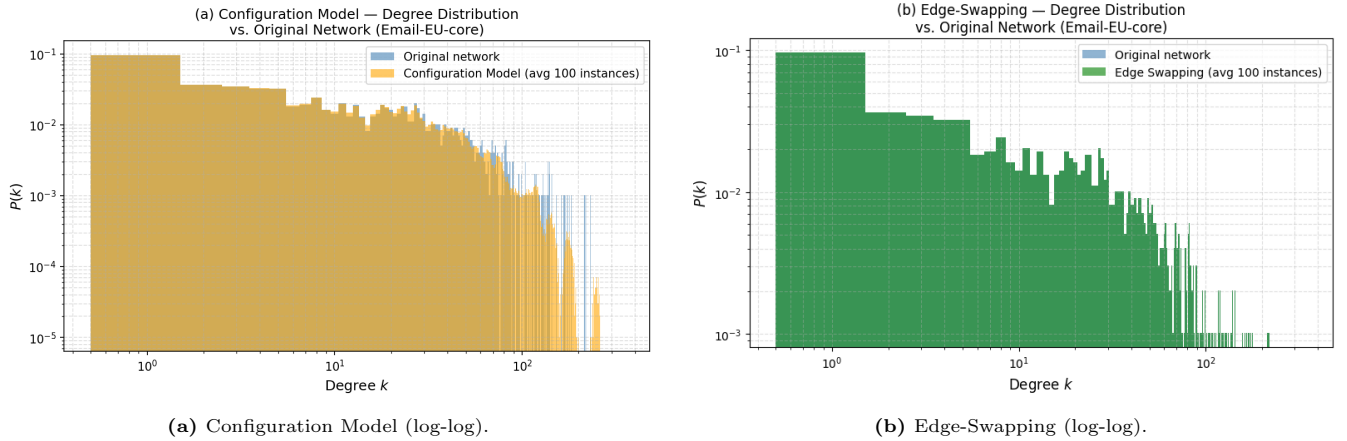


Figure 1: Averaged degree distributions (100 instances) vs. original.

Observations. Both methods reproduce the original degree distribution well. The Configuration Model shows a small high-degree deficit because high-degree nodes produce more stubs and are more likely to generate discarded self-loops or multi-edges; the mean degree from 100 CM instances is marginally below the original 32.58 due to this effect. Edge-Swapping preserves the degree sequence exactly in every instance (mean degree difference from original = 0.0000) since node degrees never change during a swap. Both serve as degree-preserving null models for studying higher-order structural properties.

2. Degree Correlations (k_{nn} vs k)

Method. For each node i , the average nearest-neighbour degree is $k_{nn}(i) = \frac{1}{k_i} \sum_{j \in \mathcal{N}(i)} k_j$. Values are binned by k and averaged to form $k_{nn}(k)$. A random-graph baseline (100 Configuration Model instances) provides the uncorrelated reference.

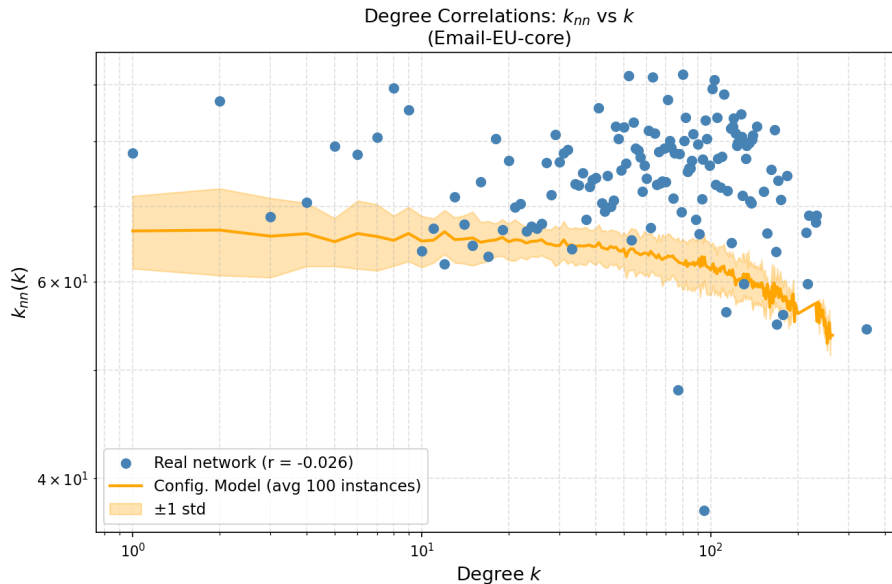


Figure 2: k_{nn} vs k for the real network and the random-graph baseline (± 1 std). Pearson $r = -0.0257$.

Observations. Pearson assortativity $r = -0.0257$, indicating a nearly uncorrelated network. The mean k_{nn} of the real network and of the random baseline (averaged over 100 Configuration Model instances) are very close, confirming that the degree sequence alone accounts for most of the apparent correlation structure. The $k_{nn}(k)$ curve is mostly flat and tracks the random baseline closely across all k ; deviations at the high-degree tail are due to the small sample of high-degree nodes.

3. Residue Interaction Graph (RIG) and Long-Range Interaction Network (LIN)

Selected proteins.

PDB ID	Protein	Organism	Structure	Residues
1UBQ	Ubiquitin	Human	Mixed α/β	76
1HRC	Cytochrome C	Horse heart	Mostly α	104
1FKB	FKBP12	Human	β -sheet rich	107
1APS	Acylphosphatase	Human	Mixed α/β	98
1CSP	Cold Shock Prot. B	<i>B. subtilis</i>	β -barrel	67

Construction. RIG: edge between residues i and j if $C\alpha-C\alpha$ distance $< 8 \text{ \AA}$ and $|i - j| \geq 2$. **LIN:** same distance cutoff, but $|i - j| > 12$ (long-range only).

Table 1: Network properties for RIG and LIN models.

Protein	RIG				LIN			
	Edges	$\langle k \rangle$	L	C	Edges	$\langle k \rangle$	L	C
1UBQ	251	6.61	3.16	0.326	90	2.37	5.41	0.000
1HRC	385	7.40	3.49	0.336	115	2.21	6.43	0.008
1FKB	433	8.09	3.35	0.346	212	3.96	4.27	0.025
1APS	392	8.00	3.16	0.327	205	4.18	4.23	0.030
1CSP	242	7.22	2.77	0.373	102	3.05	3.66	0.000

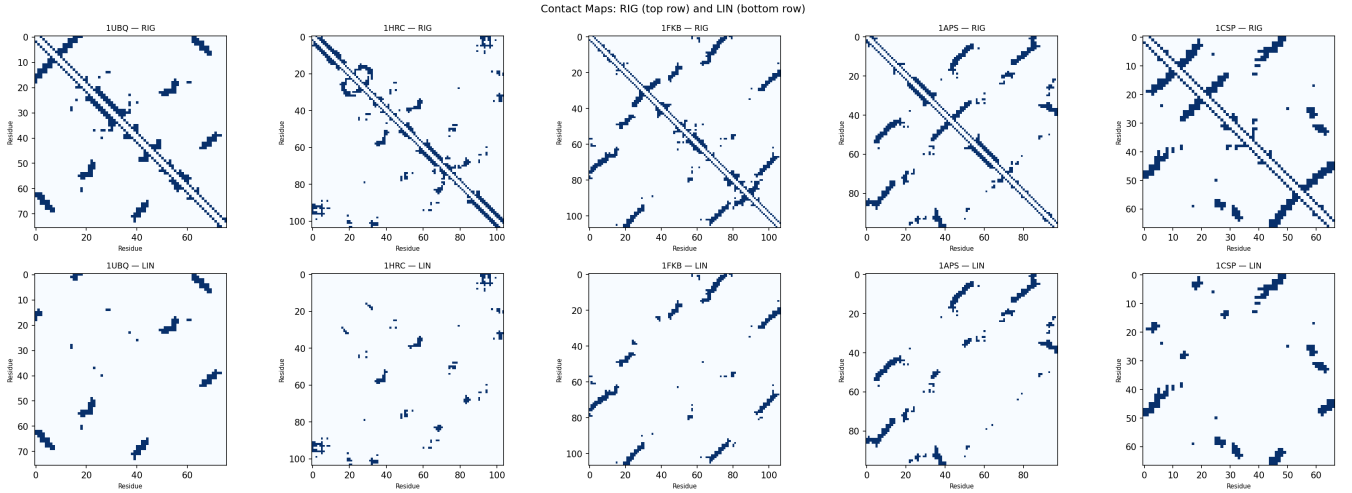


Figure 3: Contact maps: RIG (top) and LIN (bottom) for all five proteins.

Observations. All RIG networks show small-world character: $L \approx 2.8-3.5$ and $C \approx 0.33-0.37$, reflecting the dense local packing of protein structures. LIN networks are much sparser; β -sheet proteins (1FKB, 1APS) retain more LIN edges because β -sheets bring sequentially distant residues into 3D proximity. LIN clustering is near zero ($C \approx 0-0.03$) as long-range contacts rarely form triangles. Proteins with a high LIN/RIG edge ratio (1FKB, 1CSP) are expected to fold more slowly because forming long-range contacts requires a larger conformational search.

4. Bartoli’s Model of Protein Structure

Model procedure. (i) Fill the first two off-diagonals of the adjacency matrix (backbone). (ii) Sample a pair (i, j) with $P \propto 1/|i - j|$. (iii) Set all 9 entries of $\{i \pm 1\} \times \{j \pm 1\}$ to 1. (iv) Repeat until edge count reaches the target (real RIG edge count). 100 instances were run per protein.

Table 2: L and C for real structures, standard Bartoli, and modified Bartoli (mean $\pm$ std, 100 instances).

Protein	Real RIG		Real LIN		Bartoli		Mod. Bartoli	
	L	C	L	C	L	C	L	C
1UBQ	3.16	0.326	5.42	0.000	4.07 $\pm$ 0.49	0.578	5.67 $\pm$ 0.79	0.603
1HRC	3.49	0.335	6.43	0.008	4.01 $\pm$ 0.34	0.564	6.46 $\pm$ 0.96	0.604
1FKB	3.35	0.346	4.27	0.025	3.65 $\pm$ 0.27	0.561	5.97 $\pm$ 0.86	0.605
1APS	3.16	0.327	4.23	0.030	3.52 $\pm$ 0.22	0.563	5.55 $\pm$ 0.83	0.604
1CSP	2.77	0.373	3.66	0.000	3.52 $\pm$ 0.49	0.586	4.48 $\pm$ 0.59	0.603

Part (a): Comparison with real RIG and LIN. Observations. Standard Bartoli produces L values moderately above the real RIG ($\approx 3.5-4.1$ vs. real $\approx 2.8-3.5$) and significantly over-estimates clustering ($C \approx 0.56-0.59$ vs. real $\approx 0.33-0.37$). The over-clustering arises because assigning 9 contacts at once in step (iii) creates artificial triangles at a density not found in real proteins.

Modified step (ii): We tested $P(i, j) \propto \frac{1}{|i - j|} \exp\left(-\frac{(|i - j| - 8)^2}{72}\right)$, peaking at $|i - j| \approx 8$ to mimic secondary-structure contact distances. This *increased* L further from the real RIG rather than reducing it: concentrating contacts at medium range creates a more chain-like structure with longer paths. Clustering remained similarly elevated. A meaningful improvement would require incorporating residue-type and hydrophobicity information rather than sequence distance alone.

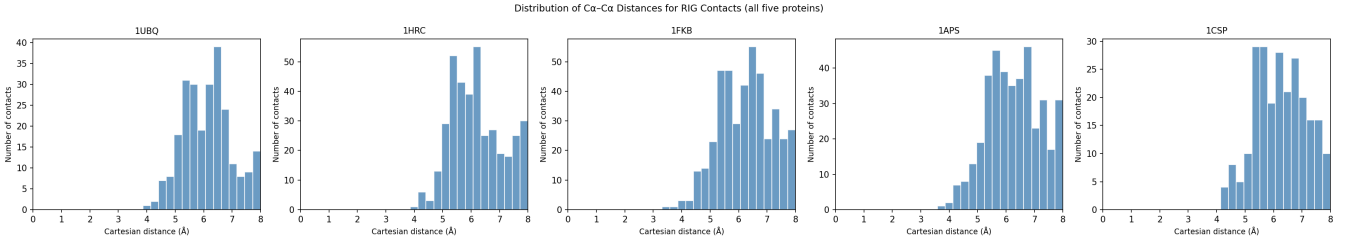


Figure 4: Distribution of $C\alpha-C\alpha$ distances across all RIG contacts for each protein (8 Å cutoff).

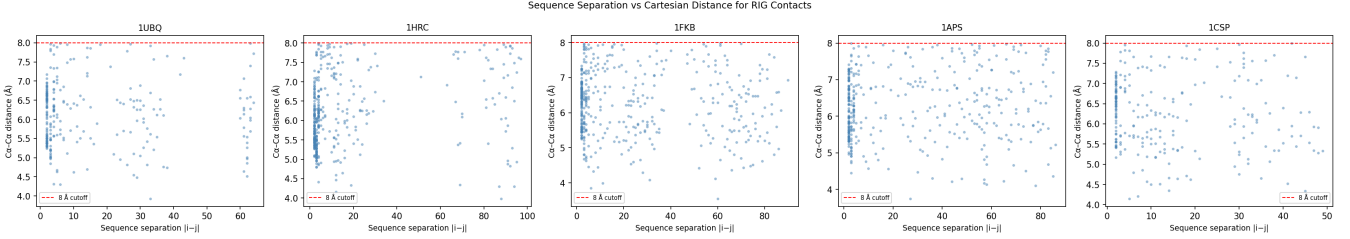


Figure 5: Sequence separation $|i - j|$ vs. 3D distance for all RIG contacts. Red dashed line marks the 8 Å cutoff.

Part (b): Contacts vs Cartesian distance. Trend 1 – distance histogram. $C\alpha-C\alpha$ distances for RIG contacts cluster heavily between 5 and 8 Å, peaking at 6–7 Å. Contacts below 4 Å are essentially absent since those distances correspond to covalent bonds already encoded in the backbone diagonals. This pattern is consistent across all five proteins.

Trend 2 – sequence separation vs. Cartesian distance. The scatter plot reveals a “funnel” structure: short-range contacts ($|i - j| < 5$) are tightly constrained at 5–7 Å by backbone geometry with almost no scatter, whereas long-range contacts ($|i - j| > 20$) spread broadly from 3.5 Å to just under the 8 Å cutoff because they are determined entirely by the tertiary fold.

Can this improve the model? Yes, concretely. (1) A *two-regime probability*: one component peaked at $|i - j| \approx 3-5$ for secondary-structure contacts and a second, flatter component for $|i - j| > 12$ for tertiary contacts – neither Bartoli variant does this. (2) *Distance-conditioned placement*: using the observed distance distributions as priors when assigning the 3×3 block, rather than ignoring Cartesian distance altogether. Both Bartoli variants fail because they select pairs based solely on sequence separation and blindly assign the 3×3 neighbourhood; without incorporating actual spatial proximity the bimodal contact structure cannot be reproduced.